

LANGLADE COUNTY AP, WI, USA

WMO: 726626

Lat: 45.154N

Lon: 89.111W

Elev: 1522

StdP: 13.91

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04864

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-9.4	-21.9	1.7	-12.6	-17.1	2.3	-9.1	24.6	17.6	22.3	16.4	5.3	280	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	84.2	71.4	81.7	69.4	79.4	67.4	74.5	80.6	72.3	78.7	70.3	76.4	10.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	127.4	77.9	70.3	118.1	75.9	68.2	109.6	73.9	39.0	81.2	37.0	78.6	35.2	76.1	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	20.0	18.0	DB	-22.7	89.4	4.5	3.5	-26.0	91.9	-28.6	93.9	-31.2	95.9	-34.4	98.4	(4)
(5)			WB	-23.0	77.2	4.4	2.8	-26.2	79.3	-28.8	80.9	-31.3	82.5	-34.5	84.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	13.6	15.6	27.1	41.1	53.2	62.4	66.6	64.6	57.8	45.1	32.4	18.9		
	DBStd	21.06	12.18	11.60	12.11	9.53	8.44	6.58	5.67	5.52	8.06	8.52	9.83	11.70		
	HDD50	4885	1129	963	716	292	62	2	0	0	21	205	529	966		
	HDD65	8767	1594	1383	1175	717	379	124	50	75	242	620	977	1431		
	CDD50	1842	0	0	6	26	160	374	514	453	254	52	3	0		
	CDD65	245	0	0	0	0	11	46	98	63	25	2	0	0		
	CDH74	2273	0	0	0	13	150	402	907	559	224	18	0	0		
	CDH80	450	0	0	0	1	27	70	208	102	42	0	0	0		
	WSAvg	8.0	8.7	8.7	8.4	9.3	8.8	7.3	6.1	5.8	7.1	8.3	8.6	8.5		
	PrecAvg	30.9	1.0	0.9	1.5	2.6	3.7	4.1	3.7	3.6	3.9	2.9	1.7	1.3		
(15) Precipitation	PrecMax	38.5	2.7	2.0	2.9	4.8	5.5	7.5	9.6	5.9	8.9	5.6	5.4	2.6		
	PrecMin	25.4	0.2	0.3	0.2	1.0	1.5	1.7	0.9	1.0	0.6	0.4	0.2	0.6		
	PrecStd	4.0	0.6	0.4	0.8	1.2	1.3	1.6	2.0	1.2	2.1	1.5	1.1	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	45.5	66.2	75.1	83.7	85.8	89.9	87.3	84.2	76.8	61.5	49.1		
		MCWB	36.0	41.4	57.6	58.6	68.2	73.7	74.6	73.0	69.9	63.2	53.7	48.0		
	2%	DB	36.1	39.1	55.2	68.6	78.7	81.7	84.4	82.3	80.7	70.0	55.0	39.5		
		MCWB	34.1	36.2	47.4	54.3	64.0	69.5	72.3	70.0	67.7	61.4	49.4	37.1		
	5%	DB	33.6	35.9	48.2	63.1	73.2	79.1	81.8	80.6	75.4	64.0	50.9	36.3		
		MCWB	32.4	33.9	43.0	50.3	60.2	67.3	70.4	68.5	65.0	56.5	46.1	34.9		
	10%	DB	30.4	32.2	43.0	57.1	68.8	75.1	79.3	76.9	72.2	59.2	45.9	33.8		
		MCWB	29.6	30.7	38.4	47.1	58.5	65.3	68.2	66.3	63.5	53.6	42.2	32.8		
	0.4%	WB	36.7	42.7	58.0	62.6	70.8	75.9	79.0	76.1	73.2	67.9	56.1	48.2		
		MCDB	39.3	44.8	64.5	71.2	79.6	81.9	84.8	82.0	79.9	71.4	59.9	48.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.1	36.7	50.1	56.4	66.4	72.9	75.1	73.1	70.3	62.1	51.2	38.4		
		MCDB	35.7	39.0	54.2	65.3	74.4	78.6	80.6	79.4	76.4	66.6	54.8	39.3		
	5%	WB	32.2	33.8	43.3	52.2	63.6	70.1	72.7	70.8	68.0	57.8	46.1	35.1		
		MCDB	33.2	35.6	47.7	60.4	69.7	76.0	79.3	76.6	73.3	63.4	48.9	36.3		
	10%	WB	29.7	30.5	38.6	48.2	60.3	67.3	70.5	68.6	65.5	54.1	42.7	32.4		
		MCDB	30.6	31.8	42.3	56.6	66.5	73.3	76.8	74.5	70.1	58.4	46.1	33.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	18.5	19.0	19.8	21.5	20.3	21.8	21.4	21.2	17.4	14.1	13.8		
		MCDBR	15.0	18.1	22.2	29.8	27.4	23.8	24.1	24.5	25.0	24.8	21.3	14.5		
		MCWBR	13.1	14.6	15.1	17.4	14.7	13.1	12.9	12.8	13.7	15.5	15.4	12.4		
	5% WB	MCDBR	14.3	16.5	20.3	26.5	22.2	20.5	20.7	20.6	20.5	21.2	19.5	13.4		
		MCWBR	12.8	13.8	15.0	17.5	13.6	13.1	13.0	12.4	13.0	15.1	15.3	11.8		
(40) Clear-Sky Solar Irradiance	taub		0.242	0.279	0.314	0.338	0.376	0.375	0.367	0.369	0.346	0.309	0.282	0.252		
	taud		2.397	2.283	2.355	2.402	2.331	2.358	2.397	2.390	2.456	2.556	2.535	2.450		
	Ebn at Noon		284	287	292	293	283	284	279	278	277	267	267	268		
	Edh at Noon		23	31	34	35	39	38	36	35	30	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		506	857	1231	1451	1715	1862	1918	1630	1282	800	489	361		
	RadStd		52	85	126	143	141	146	100	107	90	82	49	39		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page